

BPR-Net: Bin-Resolved Temporal Responses for Frequency-Aware Prototype Routing in Spatiotemporal Forecasting

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Abstract

Spatiotemporal forecasting requires modeling temporal dynamics and spatial dependencies across multiple spatial units. Existing approaches construct spatial relations from graph structures, latent representations, or attention weights. This work uses frequency-bin-level temporal responses in the prototype routing cost before spatial-unit-level aggregation. We propose BPR-Net (Bin-aware Prototype Routing Network), a frequency-bin-aware prototype routing model for spatiotemporal forecasting. BPR-Net explicitly represents frequency-bin-level temporal responses before they are aggregated into spatial-unit-level representations and uses them for prototype-based spatial relation learning. Specifically, BPR-Net represents each spatial unit through bin-resolved temporal responses, incorporates them into the prototype routing cost via Bin-Prototype Affinity, and modulates the routed representation using a spectral signature. Experiments on nine datasets across mobile traffic, air quality, taxi flow, and traffic sensor domains show that BPR-Net achieves the best result in 26 out of 28 dataset-metric settings against 19 baselines. Additional analyses show that BPR-Net achieves low forecasting error with short inference time in the reported efficiency settings and maintains low forecasting errors for spatial units with different spectral properties.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Information systems** → **Spatial-temporal systems**.

Keywords

Spatiotemporal Forecasting; Frequency-domain Learning; Prototype Routing; Spatial Relation Learning; Time Series Forecasting

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1 Introduction

Spatiotemporal forecasting has been studied in various data-driven applications, including mobile traffic analysis, air-quality monitoring, and traffic-flow prediction [18, 29, 38]. This task aims to predict future observations by modeling temporal dynamics and spatial dependencies among multiple spatial units [37]. Recent studies have investigated diverse model designs for spatiotemporal forecasting, including recurrent and convolutional architectures, adaptive spatial relation learning, attention-based modeling, graph filtering mechanisms, and retrieval-augmented designs [1, 5, 14, 18, 23, 34, 37].

Spatiotemporal datasets differ in temporal and spatial characteristics. For example, datasets show different levels of periodicity, spectral dispersion, effective frequency composition, and global spatial correlation. In addition, temporal patterns and spatial dependencies vary across spatial units within the same spatial domain [1, 7].

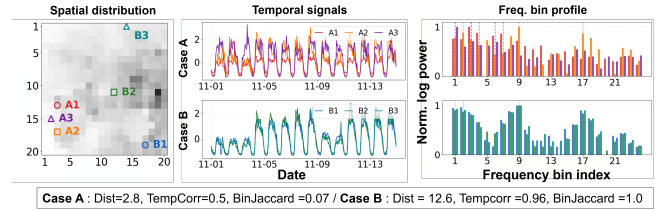


Figure 1: Case study of spatial distance and temporal-frequency similarity on MI-Internet.

Figure 1 presents two representative cases comparing spatial distance and temporal-frequency similarity among selected spatial units. Case A shows nearby cells with different temporal signals and frequency profiles, whereas Case B shows distant cells with similar temporal signals and frequency profiles. These examples indicate that spatial distance and temporal-frequency similarity are not always aligned.

These cases show that spatial distance and temporal-frequency similarity describe different properties of spatial units. Temporal-frequency similarity alone does not establish a spatial dependency between spatial units. In this work, it is used as an additional signal for spatial relation learning beyond spatial proximity. Frequency-domain representations characterize periodicity, multi-scale fluctuations, and repeated patterns in temporal signals [36, 43].

Representative spatiotemporal forecasting models construct spatial relations from graph structures, adaptive adjacency matrices, attention weights, or learned node representations [5, 14, 18, 23, 33,

37]. When learned representations are used for relation learning, spatial relations are computed from spatial-unit-level representations that contain aggregated temporal information. In these formulations, frequency-bin-level temporal responses are not used as a separate input to the relation-learning criterion. Existing frequency-domain forecasting models use spectral information for temporal representation learning or long-term dependency modeling [31, 36, 43].

BPR-Net preserves frequency-bin-level temporal responses before spatial-unit-level aggregation and uses them in prototype routing. The difference lies in how spectral information is used for spatial relation learning: BPR-Net incorporates frequency-bin-level temporal responses into the routing cost instead of using only a spatial-unit-level representation.

The contributions of this work are summarized as follows:

- We propose a Bin-Resolved Spectral Encoder that preserves frequency-bin-level temporal responses before they are aggregated into spatial-unit-level representations and provides these responses for prototype routing.
- We propose Bin-Prototype Affinity, which incorporates frequency-bin-level temporal responses into the prototype routing cost. This allows prototype assignment to use bin-level responses in addition to the spatial-unit-level representation.
- We evaluate BPR-Net on nine datasets across mobile traffic, air quality, taxi flow, and traffic sensor domains. BPR-Net achieves the best result in 26 out of 28 dataset–metric settings. Ablation and routing analyses examine the effects of frequency-bin-level temporal responses on forecasting performance and prototype routing behavior.

2 Related Work

2.1 Spatiotemporal Forecasting Models

Spatiotemporal forecasting models have been developed to model temporal dynamics and spatial dependencies in structured time-series data. Recurrent and convolutional models, such as ConvLSTM [24], STResNet [41], and STCNet [40], encode temporal evolution and local spatiotemporal patterns through sequential or convolutional operations. Graph-based models, including STGCN [37], MTGNN [33], MVSTGN [35], BigST [11], and MAGE [21], model spatial dependencies using graph structures, adaptive relations, or learned node representations. Other models, such as GLSTTN [9], TimeFilter [14], ST-ReP [42], DSFormer [25], and RAST [23], use attention weights, filtering mechanisms, representation learning, or retrieved patterns for dependency modeling.

These methods construct spatial relations from graph structures, learned adjacency matrices, node representations, attention scores, filtering weights, or retrieved patterns. When learned representations are used, temporal information is contained in spatial-unit-level representations before spatial relations are computed. In contrast, BPR-Net retains frequency-bin-level temporal responses and uses them in prototype routing before they are aggregated into spatial-unit-level representations.

2.2 Temporal Pattern and Spectral Modeling

Temporal pattern modeling has been studied through decomposition, autocorrelation, periodic modeling, and frequency-domain representation. Autoformer [32] uses series decomposition and an autocorrelation mechanism. FEDformer [43] combines decomposition with frequency-domain enhancement, and FreTS [36] learns temporal representations in the frequency domain. Recent models also use Transformer-based architectures, simplified temporal modules, or representation reformulation for temporal modeling [4, 5, 30].

These methods use temporal or spectral information to construct forecasting representations, model periodicity, or capture long-term dependencies. In these formulations, spectral information is used primarily to construct temporal representations. In BPR-Net, frequency-bin-level temporal responses are used in the prototype routing cost for spatial relation learning.

2.3 Prototype-based Routing and Alignment

Prototype-based approaches have been used to represent latent spatiotemporal structures. STPro [39] learns spatiotemporal prototypes for OD demand prediction, while ST-SSDL [6] uses learnable prototypes to discretize latent representations and model deviations between current observations and historical patterns. In these approaches, prototypes are used to represent mobility patterns, latent structures, or current-history deviations.

For spatial relation learning, prototype-based relations are often defined from latent representations, historical patterns, or current-history deviations. In these formulations, prototypes serve as latent representatives of spatiotemporal patterns, and their relations are not explicitly computed from frequency-bin-level temporal responses. In contrast, BPR-Net modifies the spatial-unit–prototype routing cost with Bin-Prototype Affinity computed from frequency-bin-level temporal responses.

3 Methodology

3.1 Problem Formulation

We consider next-step spatiotemporal forecasting over a set of N spatial units, where each unit represents either a graph node or a flattened grid cell. Given a historical input tensor $\mathbf{X} \in \mathbb{R}^{P \times N \times C}$, where P is the input sequence length and C is the number of channels, the objective is to predict the next-step target $\mathbf{Y} \in \mathbb{R}^{N \times C}$.

Formally, the forecasting problem is defined as learning a mapping function f_θ such that

$$\hat{\mathbf{Y}} = f_\theta(\mathbf{X}), \quad \hat{\mathbf{Y}} \in \mathbb{R}^{N \times C}, \quad (1)$$

where θ denotes the learnable parameters of the model.

3.2 Bin-Resolved Spectral Encoder (BRSE)

Existing frequency-domain temporal encoders use spectral information to construct temporal representations [36, 43]. Fourier analysis represents a temporal signal as coefficients associated with different frequency bases [22]. When these coefficients are transformed by a shared mapping, the same transformation rule is applied to different frequency bins. In spatiotemporal data, different frequency ranges correspond to different temporal behaviors across spatial units.

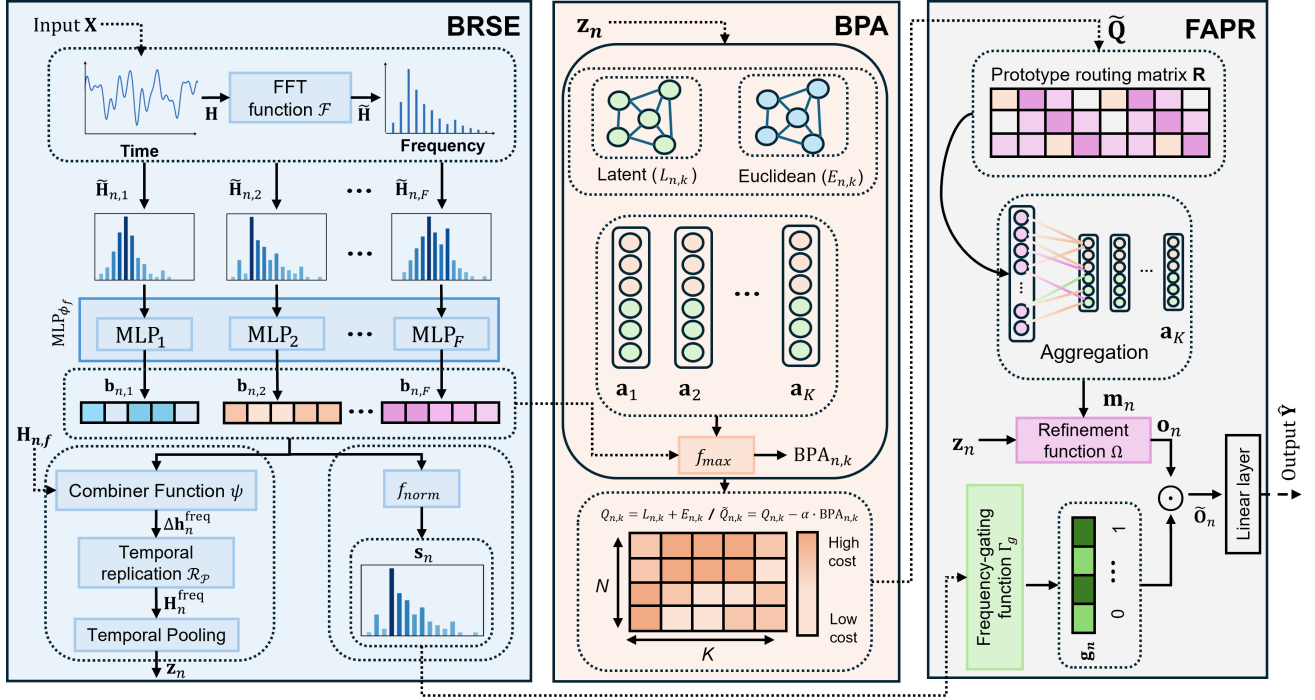


Figure 2: Overview of frequency-bin-aware prototype routing in BPR-Net.

We use a Bin-Resolved Spectral Encoder (BRSE) to keep frequency-bin-level responses explicit and to apply a separate transformation to each selected frequency bin. The resulting bin-level features are used to compute Bin-Prototype Affinity for prototype-based spatial relation learning. Figure 2 illustrates how frequency-bin-level responses are obtained and used in prototype routing.

The input tensor is first projected to a hidden representation $\mathbf{H} \in \mathbb{R}^{P \times N \times D}$ by a linear input projection, where D is the hidden dimension. We then apply the Fast Fourier Transform along the temporal dimension for each spatial unit:

$$\tilde{\mathbf{H}}_n = \mathcal{F}(\mathbf{H}_{:,n,:}) \in \mathbb{C}^{F \times D}, \quad (2)$$

where $\mathcal{F}(\cdot)$ denotes the Fast Fourier Transform along the temporal dimension, and F denotes the number of frequency bins retained after the transform. Each coefficient $\tilde{\mathbf{H}}_{n,f}$ represents the response of spatial unit n to the f -th temporal frequency basis, yielding a set of frequency-bin-level components.

For each frequency bin $f = 1, \dots, F$, the real and imaginary parts of the complex frequency representation are concatenated and transformed by a frequency-wise MLP ϕ_f :

$$\mathbf{b}_{n,f} = \phi_f \left(\left[\text{Re}(\tilde{\mathbf{H}}_{n,f}); \text{Im}(\tilde{\mathbf{H}}_{n,f}) \right] \right) \in \mathbb{R}^D, \quad (3)$$

where $\mathbf{b}_{n,f}$ denotes the feature vector of the f -th frequency bin for spatial unit n , and $\phi_f : \mathbb{R}^{2D} \rightarrow \mathbb{R}^D$ denotes the frequency-wise MLP for the f -th bin. The transformations $\{\phi_f\}_{f=1}^F$ are parameterized independently. Thus, frequency bins are transformed by bin-specific mappings rather than a single shared mapping. This parameterization keeps bin-dependent responses available for spatial relation learning.

Let $\mathcal{G}_{\text{shared}}$ denote the class of frequency-wise mappings with a shared transformation ϕ for all bins, and let $\mathcal{G}_{\text{bin}}$ denote the class of mappings with independently parameterized transformations $\{\phi_f\}_{f=1}^F$. Since the shared case is obtained by setting $\phi_1 = \dots = \phi_F = \phi$, we have

$$\mathcal{G}_{\text{shared}} \subseteq \mathcal{G}_{\text{bin}}. \quad (4)$$

This inclusion shows that the shared transformation is a restricted case of the bin-wise formulation. This parameterization allows frequency-dependent transformations that cannot be represented when all frequency bins share the same mapping.

The resulting bin features are stacked to form the spatial-unit-wise bin feature matrix $\mathbf{B}_n \in \mathbb{R}^{F \times D}$. The bin feature matrices of all spatial units are collected as $\mathbf{B} \in \mathbb{R}^{N \times F \times D}$. This tensor is later used to compute Bin-Prototype Affinity.

The bin features are then concatenated and passed through a combiner function $\psi : \mathbb{R}^{FD} \rightarrow \mathbb{R}^D$ to obtain a spatial-unit-level frequency correction:

$$\Delta \mathbf{h}_n^{\text{freq}} = \psi([\mathbf{b}_{n,1}; \dots; \mathbf{b}_{n,F}]) \in \mathbb{R}^D. \quad (5)$$

This correction is expanded along the temporal dimension through a broadcasting function and added to the hidden representation:

$$\mathbf{H}_{:,n,:}^{\text{freq}} = \mathbf{H}_{:,n,:} + \lambda \mathcal{R}_P(\Delta \mathbf{h}_n^{\text{freq}}), \quad (6)$$

where $\mathcal{R}_P(\cdot) : \mathbb{R}^D \rightarrow \mathbb{R}^{P \times D}$ denotes temporal replication over P time steps, and $\lambda = 0.3$ controls the spectral correction. The correction term is shared across the temporal positions of the same spatial unit. Therefore, it does not replace the original temporal hidden sequence, but injects a spatial-unit-level summary of its spectral composition into the hidden representation.

Prototype routing requires one representation for each spatial unit. Therefore, we aggregate the frequency-corrected hidden sequence over the temporal dimension using learnable temporal pooling weights:

$$\mathbf{z}_n = \sum_{p=1}^P \beta_p \mathbf{H}_{p,n,:}^{\text{freq}} \in \mathbb{R}^D, \quad (7)$$

where $\boldsymbol{\beta} = [\beta_1, \dots, \beta_P] \in \mathbb{R}^P$ denotes learnable normalized pooling weights over the temporal dimension. The spatial-unit-level representations are collected as

$$\mathbf{Z} = [\mathbf{z}_1; \dots; \mathbf{z}_N] \in \mathbb{R}^{N \times D}. \quad (8)$$

We also compute a spectral signature from the same frequency-bin features. For each spatial unit n and frequency bin f , we compute the magnitude of the transformed bin feature:

$$e_{n,f} = \|\mathbf{b}_{n,f}\|_2. \quad (9)$$

The spectral signature is obtained by applying a normalization function over the frequency-bin dimension:

$$\mathbf{s}_n = f_{\text{norm}}([e_{n,1}, \dots, e_{n,F}]) \in \mathbb{R}^F, \quad (10)$$

where $f_{\text{norm}}(\cdot)$ denotes ℓ_2 -normalization over the frequency-bin dimension. Thus, $\mathbf{s}_n$ summarizes the relative activation strength of the transformed frequency-bin features for spatial unit n , rather than the raw Fourier energy of the original signal. The spectral signatures of all spatial units are collected as $\mathbf{S} = [\mathbf{s}_1; \dots; \mathbf{s}_N] \in \mathbb{R}^{N \times F}$.

3.3 Bin-Prototype Affinity (BPA)

Existing prototype-based methods use prototypes to represent spatiotemporal patterns, latent structures, or current-history deviations, rather than to compute routing costs from frequency-bin-level temporal responses [6, 39]. To incorporate frequency-bin-level responses into the routing cost, we compute a Bin-Prototype Affinity (BPA) score between the bin feature tensor $\mathbf{B}$ and a set of learnable prototypes.

Let $\mathbf{A} = [\mathbf{a}_1; \dots; \mathbf{a}_K] \in \mathbb{R}^{K \times D}$ denote the learnable prototype matrix, where $\mathbf{a}_k$ is the k -th prototype vector. The prototypes are used in both the base routing cost and Bin-Prototype Affinity. They do not represent predefined spatial regions or fixed frequency patterns. The prototype vectors are initialized from a zero-mean Gaussian distribution with a standard deviation of 0.02 and are optimized during training.

Before incorporating frequency-bin-level responses, we first compute a base routing cost using latent representation distance and Euclidean distance. For the latent representation distance, we apply linear projections $\mathbf{q}_n = \mathbf{W}_q \mathbf{z}_n$ and $\mathbf{p}_k = \mathbf{W}_p \mathbf{a}_k$. The latent representation distance is defined as

$$L_{n,k} = 1 - \langle \hat{\mathbf{q}}_n, \hat{\mathbf{p}}_k \rangle, \quad (11)$$

where $\hat{\mathbf{q}}_n$ and $\hat{\mathbf{p}}_k$ denote the normalized projected spatial-unit and prototype representations, respectively. The optional Euclidean distance term is computed in the coordinate space. Let $\mathbf{r}_n \in \mathbb{R}^2$ denote the coordinate of spatial unit n , and let $\boldsymbol{\rho}_k \in \mathbb{R}^2$ denote the learnable coordinate of prototype k . We define

$$E_{n,k} = \|\mathbf{r}_n - \boldsymbol{\rho}_k\|_2. \quad (12)$$

When coordinate-based routing is not used, $E_{n,k}$ is set to zero. The base routing cost is defined as

$$Q_{n,k} = L_{n,k} + E_{n,k}. \quad (13)$$

The spatial coordinates are normalized to a common range before computing the Euclidean routing cost.

To incorporate frequency-bin-level responses into routing, we compute a Bin-Prototype Affinity (BPA) score between the bin feature tensor $\mathbf{B}$ and the prototype matrix $\mathbf{A}$. Before computing similarity, both bin features and prototype vectors are ℓ_2 -normalized along the feature dimension. The affinity between spatial unit n and prototype k is computed by applying a pooling function over frequency-bin-wise cosine similarities:

$$\text{BPA}_{n,k} = f_{\text{max}}\left(\left\{\langle \hat{\mathbf{b}}_{n,f}, \hat{\mathbf{a}}_k \rangle\right\}_{f=1}^F\right), \quad (14)$$

where $f_{\text{max}}(\cdot)$ denotes max pooling over the frequency-bin dimension, and $\hat{\mathbf{b}}_{n,f}$ and $\hat{\mathbf{a}}_k$ denote the normalized bin feature vector and prototype vector, respectively. This value represents the largest frequency-bin-level similarity between spatial unit n and prototype k . We use max pooling to match each prototype with the strongest frequency-bin response rather than the average spectral profile of a spatial unit. This avoids averaging frequency-bin responses before computing the spatial-unit-prototype affinity.

The BPA score defines a set-to-prototype matching criterion. For spatial unit n , define the set of normalized frequency-bin features as

$$\mathcal{B}_n = \{\hat{\mathbf{b}}_{n,1}, \dots, \hat{\mathbf{b}}_{n,F}\}. \quad (15)$$

Since $\mathcal{B}_n$ is a finite set of frequency-bin features, Eq. (14) is written as

$$\text{BPA}_{n,k} = \max_{\mathbf{u} \in \mathcal{B}_n} \langle \mathbf{u}, \hat{\mathbf{a}}_k \rangle. \quad (16)$$

Thus, BPA measures the strongest alignment between prototype k and the frequency-bin-level components of spatial unit n . Unlike a routing criterion based only on the aggregated representation $\mathbf{z}_n$, this formulation uses the set of bin-level responses $\mathcal{B}_n$ and allows a prototype to be matched with the frequency component that gives the maximum cosine similarity.

The BPA score is incorporated into the routing cost as a frequency-aware correction term:

$$\tilde{Q}_{n,k} = Q_{n,k} - \alpha \cdot \text{BPA}_{n,k}, \quad (17)$$

where $\alpha > 0$ is a learnable scalar that controls the contribution of BPA and is implemented with a positivity constraint, such as a softplus parameterization. A larger BPA score decreases the routing cost between spatial unit n and prototype k .

This formulation gives BPA a direct monotonic effect on the routing cost. Since

$$\frac{\partial \tilde{Q}_{n,k}}{\partial \text{BPA}_{n,k}} = -\alpha < 0, \quad (18)$$

a larger BPA score reduces the frequency-aware routing cost between spatial unit n and prototype k when $\alpha > 0$.

The additional BPA computation requires $O(NKFD)$ operations for N spatial units, K prototypes, F frequency bins, and hidden dimension D .

3.4 Frequency-Aware Prototype Routing (FAPR)

Given the frequency-aware routing cost matrix $\tilde{\mathbf{Q}} \in \mathbb{R}^{N \times K}$, we compute the spatial-unit-prototype routing weight matrix $\mathbf{R} \in \mathbb{R}^{N \times K}$ using a temperature-scaled softmax function, following common soft-assignment and attention-style weighting mechanisms [17, 28]:

$$R_{n,k} = \frac{\exp(-\tau \tilde{Q}_{n,k} / \varepsilon)}{\sum_{k'=1}^K \exp(-\tau \tilde{Q}_{n,k'} / \varepsilon)}, \quad (19)$$

where $\tau > 0$ and $\varepsilon > 0$ are routing-scale hyperparameters whose ratio τ/ε controls the sharpness of the routing distribution. The negative cost is used so that lower costs correspond to larger routing weights.

Because $\tilde{Q}_{n,k}$ includes the BPA term in Eq. (17), the routing weight is directly affected by frequency-bin-level affinity. For a fixed spatial unit n , when $\tau > 0$, $\varepsilon > 0$, and $\alpha > 0$, the derivative of the log routing weight with respect to $\text{BPA}_{n,k}$ is given by

$$\frac{\partial \log R_{n,k}}{\partial \text{BPA}_{n,k}} = \frac{\tau \alpha}{\varepsilon} (1 - R_{n,k}) \geq 0. \quad (20)$$

Thus, increasing the BPA score for prototype k increases the routing preference toward that prototype, while the normalization across prototypes is preserved by the softmax function. This relation describes the effect of frequency-bin-level affinity on the spatial-unit-prototype routing weights.

Each spatial unit aggregates the projected prototype representations using the routing weights. Let $\mathbf{P} = [\mathbf{p}_1; \dots; \mathbf{p}_K] \in \mathbb{R}^{K \times D}$ denote the matrix of projected prototype representations:

$$\mathbf{m}_n = \sum_{k=1}^K R_{n,k} \mathbf{p}_k \in \mathbb{R}^D. \quad (21)$$

The prototype-based representations of all spatial units are collected as

$$\mathbf{M} = [\mathbf{m}_1; \dots; \mathbf{m}_N] = \mathbf{R} \mathbf{P} \in \mathbb{R}^{N \times D}. \quad (22)$$

This representation is combined with the spatial-unit representation obtained from temporal encoding. Let $\mathbf{z}_n$ denote the n -th spatial-unit representation in $\mathbf{Z}$. The combination is expressed using a refinement function:

$$\mathbf{o}_n = \Omega(\mathbf{z}_n + \mathbf{m}_n), \quad (23)$$

where $\Omega(\cdot)$ denotes a refinement function consisting of layer normalization and feed-forward transformation.

Next, we compute a frequency-aware gate using the spectral signature $\mathbf{s}_n$ obtained from BRSE. Since $\mathbf{s}_n$ summarizes the relative activation strength of frequency-bin features for spatial unit n , it is used to adjust the routed representation according to the spatial-unit-level frequency activation pattern:

$$\mathbf{g}_n = \Gamma_g(\mathbf{s}_n) \in (0, 1)^D, \quad (24)$$

where $\Gamma_g(\cdot)$ denotes a frequency-gating function implemented by a linear transformation followed by a sigmoid activation, analogous to channel-wise feature recalibration mechanisms [13]. It maps the spectral signature vector to a D -dimensional gate vector. This gate is applied as an element-wise scaling factor:

$$\tilde{\mathbf{o}}_n = \mathbf{o}_n \odot \mathbf{g}_n, \quad (25)$$

where $\odot$ denotes element-wise multiplication.

After frequency-aware gating, the final routed representation matrix is written as

$$\tilde{\mathbf{O}} = [\tilde{\mathbf{o}}_1; \dots; \tilde{\mathbf{o}}_N] \in \mathbb{R}^{N \times D}. \quad (26)$$

This representation is projected to the target space through a linear output layer with a residual connection to the last observation:

$$\hat{\mathbf{Y}} = \mathbf{X}_{P,:} + \tilde{\mathbf{O}} \mathbf{W}_{\text{out}}, \quad (27)$$

where $\mathbf{X}_{P,:}$ is the last observation and $\mathbf{W}_{\text{out}} \in \mathbb{R}^{D \times C}$ is the output projection.

4 Experiments

Table 1: Summary of datasets used in the experiments.

Dataset	Domain	Size	Time Span
Milan	Mobile	20×20 crop	Nov. 2013–Jan. 2014
Trentino	Mobile	20×20 crop	Nov. 2013–Jan. 2014
PM2.5	Air quality	20×20	Jan.–Dec. 2024
PM10	Air quality	20×20	Jan.–Dec. 2024
Beijing Taxi16	Traffic	32×32	Nov. 2015–Apr. 2016
PEMS03	Traffic	358 nodes	Sep.–Nov. 2018
PEMS04	Traffic	307 nodes	Jan.–Feb. 2018
PEMS07	Traffic	883 nodes	May–Aug. 2017
PEMS08	Traffic	170 nodes	Jul.–Aug. 2016

4.1 Datasets

Table 1 summarizes the datasets used in the experiments. The datasets include four domains: mobile traffic, air quality, taxi flow, and traffic sensor networks. Milan (MI) [2, 40], Trentino (TN) [2, 40], Particulate Matter datasets (PM2.5 and PM10) [16], and Beijing Taxi16 (BJT16) [41] are represented as grid-based spatial fields, while the PEMS datasets [10] are represented as node-based sensor networks. All datasets are preprocessed to a temporal resolution of one hour per step, except PM, which retains its native 3-hour temporal resolution.

Table 2: Temporal, spectral, and spatial characteristics of the datasets.

Dataset	Periodicity	Spectral Entropy	Eff. Freq. Count	Global Spatial Corr.
MI	0.912	0.396	4.96	0.662
TN	0.952	0.445	7.21	0.581
PM2.5	0.629	0.686	5.55	0.259
PM10	0.637	0.692	5.48	0.246
BJT16	0.707	0.689	9.95	0.724
PEMS03	0.932	0.214	4.84	0.869
PEMS04	0.933	0.246	6.73	0.806
PEMS07	0.926	0.247	8.79	0.828
PEMS08	0.925	0.245	4.48	0.817

4.2 Dataset Characteristics

We analyze four dataset properties: periodicity, spectral entropy, effective frequency count, and global spatial correlation. Periodicity is measured by 24-step autocorrelation and describes daily recurrence [15, 20]. Spectral entropy measures whether temporal signal

Table 3: Forecasting results on grid-based datasets. The best results are in red bold and the second-best are underlined.

Model	MI						TN						PM		PM		BJT16			
	SMS		Call		Internet		SMS		Call		Internet		2.5		10		In		Out	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
LSTM	97.57	62.11	69.49	37.71	411.12	278.92	40.09	14.46	28.94	7.71	150.88	54.66	23.64	14.35	33.21	20.25	47.78	27.31	48.00	27.31
ConvLSTM	110.28	66.03	134.67	79.21	736.19	509.04	38.85	15.48	31.90	11.56	176.63	69.69	13.40	7.54	18.60	10.59	63.19	31.26	73.75	37.28
STResNet	68.92	38.63	44.31	24.91	351.89	225.37	51.85	27.46	28.09	14.72	216.89	118.36	28.50	18.45	40.57	26.58	53.66	35.91	52.78	35.31
STCNet	56.97	32.06	38.71	21.92	170.45	96.50	35.94	12.30	24.58	6.37	142.28	39.73	16.68	8.37	23.49	11.99	41.80	22.84	41.47	22.57
SimVP	42.67	23.00	25.75	15.15	171.11	111.98	47.39	18.05	47.92	36.51	147.77	44.63	11.34	6.59	15.88	9.31	39.89	25.52	40.73	25.77
TAU	46.77	25.20	26.98	16.16	156.23	97.13	28.38	12.56	15.14	5.76	113.79	42.77	10.91	6.25	14.48	8.11	35.19	21.81	35.09	22.13
STGCN	47.80	28.54	30.19	18.88	209.80	158.08	32.29	15.59	23.93	10.59	148.63	61.73	21.43	13.81	30.19	19.35	38.95	24.16	39.10	24.31
MTGNN	50.40	25.31	37.14	18.32	171.33	103.46	25.83	8.96	16.83	4.54	113.18	40.96	11.75	6.07	16.56	9.32	38.87	20.54	39.80	20.74
MVSTGN	46.02	25.02	25.07	14.35	164.84	107.10	63.79	53.71	43.00	34.97	199.42	181.08	13.18	10.11	19.71	15.46	41.91	25.89	40.75	24.67
BigST	45.95	27.07	40.25	22.01	181.59	115.74	35.10	15.41	25.79	7.18	116.82	59.70	14.94	8.54	20.96	12.08	39.06	21.45	43.23	24.20
GLSTTN	42.13	22.52	26.17	13.58	144.37	84.36	20.17	5.48	10.26	2.74	65.35	19.65	34.58	18.80	49.67	27.35	34.75	18.23	34.68	18.37
TimeXer	44.90	22.17	25.07	<u>12.85</u>	143.20	78.53	20.56	7.10	11.44	3.58	56.42	19.75	10.58	5.74	14.92	8.17	36.98	20.85	37.24	21.01
DRFormer	43.32	<u>21.07</u>	25.13	13.16	145.81	79.83	<u>16.41</u>	<u>5.20</u>	<u>10.17</u>	<u>2.43</u>	43.47	<u>12.65</u>	<u>10.21</u>	<u>5.37</u>	<u>14.45</u>	<u>7.66</u>	37.58	21.54	37.79	21.69
SimpleTM	41.46	21.71	27.45	14.19	155.91	91.84	27.03	15.29	19.20	11.61	86.06	56.15	12.56	6.94	17.67	9.93	41.74	23.58	41.85	23.65
ST-ReP	58.52	40.94	51.05	37.09	298.80	213.34	63.76	50.05	41.35	30.92	198.84	153.53	18.16	12.21	24.96	17.11	55.63	36.65	53.04	33.94
DSFormer	79.33	51.04	67.73	46.87	309.52	212.22	69.19	37.64	49.06	24.73	242.77	132.92	27.75	16.06	38.90	22.59	79.32	49.69	79.02	49.42
TimeFilter	<u>40.69</u>	20.85	<u>24.97</u>	13.82	<u>142.40</u>	<u>78.00</u>	17.89	5.84	11.38	3.06	46.32	16.02	10.65	5.61	15.01	7.98	OOM	OOM	OOM	OOM
MAGE	42.49	27.17	32.02	21.02	<u>175.77</u>	105.10	45.43	22.30	27.31	9.19	147.66	101.08	13.95	8.26	19.64	11.61	39.96	22.26	42.14	24.08
RAST	46.70	24.80	31.09	17.46	174.64	103.93	26.12	10.25	18.74	5.39	87.80	33.19	12.51	6.68	16.82	8.85	40.15	21.39	40.35	21.52
BPR-Net	38.87	21.89	23.17	12.23	134.56	76.57	16.02	5.05	8.37	2.27	<u>43.91</u>	12.06	9.91	5.18	13.99	7.26	33.37	18.14	33.42	18.14

Table 4: Forecasting results on node-based PEMS datasets.

Model	PEMS03		PEMS04		PEMS07		PEMS08	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
STGCN	253.33	157.78	295.36	200.17	340.69	226.12	241.76	164.04
MTGNN	249.49	146.71	<u>284.61</u>	181.11	340.01	225.76	261.21	158.76
BigST	592.97	445.71	652.09	434.67	977.96	770.00	679.36	555.46
TimeXer	278.48	179.85	330.97	225.51	373.75	262.49	280.60	189.56
DRFormer	<u>231.56</u>	<u>141.68</u>	287.94	186.32	<u>321.25</u>	208.66	<u>238.69</u>	155.14
ST-ReP	480.54	354.89	523.74	360.88	627.92	466.46	477.67	372.40
SimpleTM	275.66	163.82	317.18	207.30	371.14	251.45	276.37	180.84
DSFormer	542.08	376.06	643.30	449.90	947.22	717.68	680.09	520.73
TimeFilter	288.17	173.95	351.17	225.62	OOM	278.98	180.19	
MAGE	336.96	213.53	376.69	251.46	458.77	312.05	363.60	250.68
RAST	244.78	145.82	287.94	<u>178.29</u>	326.80	<u>208.48</u>	250.41	<u>153.59</u>
BPR-Net	214.66	123.08	252.95	157.55	282.31	172.41	206.64	122.32

energy is concentrated in a small number of frequency components or distributed across multiple frequency components [31]. Effective frequency count summarizes the approximate number of frequency components contributing to each spatial unit [12]. Global spatial correlation measures the average temporal co-variation among spatial units [18].

Table 2 shows substantial variation across the datasets. MI, TN, and the PEMS datasets exhibit high periodicity, whereas PM2.5, PM10, and BJT16 show higher spectral entropy. BJT16, TN, PEMS04, and PEMS07 also have relatively large effective frequency counts, indicating richer multi-component frequency structure in these datasets.

Spatial characteristics also vary considerably. The PEMS datasets and BJT16 show high global spatial correlation, whereas the PM datasets show lower correlation. Overall, the evaluated datasets span diverse temporal-frequency patterns and levels of spatial co-variation.

4.3 Experimental Settings

All experiments are formulated as one-step-ahead spatiotemporal forecasting. Each model predicts the next time step from the past $P = 25$ observations. We use $F = 13$ retained frequency bins, $K = 128$ learnable prototypes, and a hidden dimension of $D = 128$. BPR-Net is trained with AdamW using a learning rate of 5×10^{-4} and a batch size of 32. The training data of each dataset are chronologically split into training and validation sets at a 9:1 ratio, and the test set is used only for final evaluation.

All experiments are conducted on a single NVIDIA A100 GPU. For fair comparison, all baselines use the same input length, data split, preprocessing procedure, and evaluation protocol. Model-specific hyperparameters are set based on the original papers or official implementations when available. The source code is available at <https://github.com/D-AI-network/bprnet>.

4.4 Baselines

We compare BPR-Net with 19 baselines from major spatiotemporal forecasting model groups. The RNN-based baselines are LSTM [27] and ConvLSTM [24]. The CNN-based baselines are STResNet [41], STCNet [40], SimVP [8], and TAU [26]. The GNN-based and adaptive graph baselines are STGCN [37], MTGNN [33], MVSTGN [35], BigST [11], TimeFilter [14], and MAGE [21]. The Transformer-based baselines are GLSTTN [9], TimeXer [30], DRFormer [5], and DSFormer [25]. The remaining baselines are ST-ReP [42], RAST [23], and SimpleTM [4].

5 Results

5.1 Overall Forecasting Performance

Tables 3 and 4 report the forecasting results on grid-based and node-based datasets. BPR-Net achieves the best result in 26 out of 28 dataset-metric settings. On the mobile traffic datasets, BPR-Net

obtains the best result in most MI and TN settings across SMS, Call, and Internet traffic, which have different activity scales.

On the air-quality and transportation datasets, BPR-Net reports the lowest RMSE and MAE on PM2.5, PM10, BJT16, and all PEMS datasets. The results on PM2.5, PM10, and BJT16 are consistent with their relatively high spectral entropy, providing settings for examining frequency-bin-level temporal responses. The results on PEMS datasets, which show high global spatial correlation, provide settings for examining prototype-based spatial relation learning.

Overall, BPR-Net reports lower forecasting errors across datasets with different temporal, spectral, and spatial characteristics. These results provide the basis for the following ablation and routing analyses, which examine how frequency-bin-level responses are used in temporal encoding, routing cost computation, and frequency-aware gating.

5.2 Ablation Study

We conduct ablation experiments to examine how frequency-bin-level responses are used in BPR-Net. The experiments are performed on MI-Internet, TN-Internet, PM2.5, and PEMS08. Table 5 reports RMSE and MAE for each variant.

The w/o BRSE variant removes the bin-resolved transformation of temporal responses, eliminating the source of the bin-level features used by both Bin-Prototype Affinity and spectral-signature-based gating. The w/o BPA variant retains BRSE but removes Bin-Prototype Affinity from the routing cost, separating frequency-bin-level temporal encoding from its use in prototype routing. The Aggregated Spectral variant combines the frequency-bin-level responses before computing prototype affinity, whereas the full model computes prototype affinity from the individual bin-level responses before aggregation. The w/o Spectral Sig., Shuffled Sig., and Shared Sig. variants examine the role of spatial-unit-specific spectral signatures in frequency-aware gating.

Table 5: Ablation study on representative datasets.

Variant	MI-Internet		TN-Internet		PM2.5		PEMS08	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
w/o BRSE	209.41	140.31	63.68	16.60	15.15	6.95	440.06	294.57
w/o BPA	153.05	86.68	45.80	12.99	10.05	5.40	206.82	123.12
w/o Spectral Sig.	137.16	83.01	48.56	15.51	10.15	5.47	207.34	123.62
Shuffled Sig.	136.57	76.84	47.84	12.82	9.92	5.19	208.37	127.04
Shared Sig.	139.02	77.85	47.44	16.45	10.01	5.49	207.64	127.08
Aggregated Spectral	140.91	78.00	44.91	13.96	10.03	5.41	208.57	124.00
BPR-Net	134.56	76.57	43.91	12.06	9.91	5.18	206.64	122.32

Table 5 shows that removing BRSE increases the forecasting error across all four datasets. Since w/o BRSE removes the bin-level features used by both BPA and spectral-signature-based gating, the result shows the role of bin-resolved temporal responses as inputs to the subsequent routing and modulation components.

The w/o BPA variant also yields higher errors than the full model while retaining BRSE. Thus, extracting frequency-bin-level responses alone does not account for the full performance gain; incorporating these responses into the spatial-unit-prototype routing cost also contributes to forecasting performance.

The Aggregated Spectral variant yields higher RMSE and MAE than the full model on all four datasets. This variant retains the same frequency-bin-level responses but combines them before prototype affinity is computed. The results support computing prototype affinity from individual bin-level responses before aggregation rather than combining the responses beforehand.

The spectral-signature variants generally produce higher errors than the full model. Removing, shuffling, or sharing the spectral signature changes the spatial-unit-specific frequency information used after routing. These results support the use of the spectral signature for frequency-aware modulation of the routed representation.

Overall, the ablation results support the connected design of BPR-Net, in which BRSE constructs frequency-bin-level responses, BPA uses these responses before aggregation for prototype routing, and the spectral signature modulates the routed representation.

Table 6: Average relative error reduction of the full frequency-bin setting compared with smaller settings.

Dataset	$F = 1 \rightarrow Full$	$F = 2 \rightarrow Full$	$F = 8 \rightarrow Full$
MI-Internet	14.12%	17.92%	2.98%
TN-Internet	33.91%	5.28%	4.04%
PM2.5	29.90%	29.12%	6.30%
PM10	30.17%	29.46%	7.49%
BJT16	29.64%	22.85%	7.89%
PEMS03	41.16%	32.26%	4.18%
PEMS04	43.79%	36.07%	3.57%
PEMS07	45.23%	37.98%	3.29%
PEMS08	45.02%	36.71%	4.07%

5.3 Frequency-bin Resolution Analysis

We examine how the number of retained frequency bins affects forecasting error. This analysis differs from the component ablation because BRSE is used in all settings, while only the frequency-bin resolution is varied. Table 6 reports the average relative error reduction of the full frequency-bin setting compared with smaller frequency-bin settings, computed using RMSE and MAE for each dataset.

As shown in Table 6, the full-bin setting yields lower average errors than the reduced-bin settings across all datasets. The error reductions are larger relative to coarse resolutions, such as $F = 1$ and $F = 2$, whereas the reductions relative to $F = 8$ are smaller. These results show that retaining bin-resolved temporal responses at a sufficient frequency resolution contributes to forecasting performance within the same BRSE structure.

The dataset characteristics in Table 2 provide additional context for this result. PM2.5, PM10, and BJT16 show relatively high spectral entropy, while BJT16, TN, PEMS04, and PEMS07 show relatively large effective frequency counts [3, 22]. These properties indicate that temporal signal energy is distributed across multiple frequency components in several datasets [32]. The results are consistent with the role of BRSE, which preserves frequency-bin-level temporal responses before they are used in routing and gating.

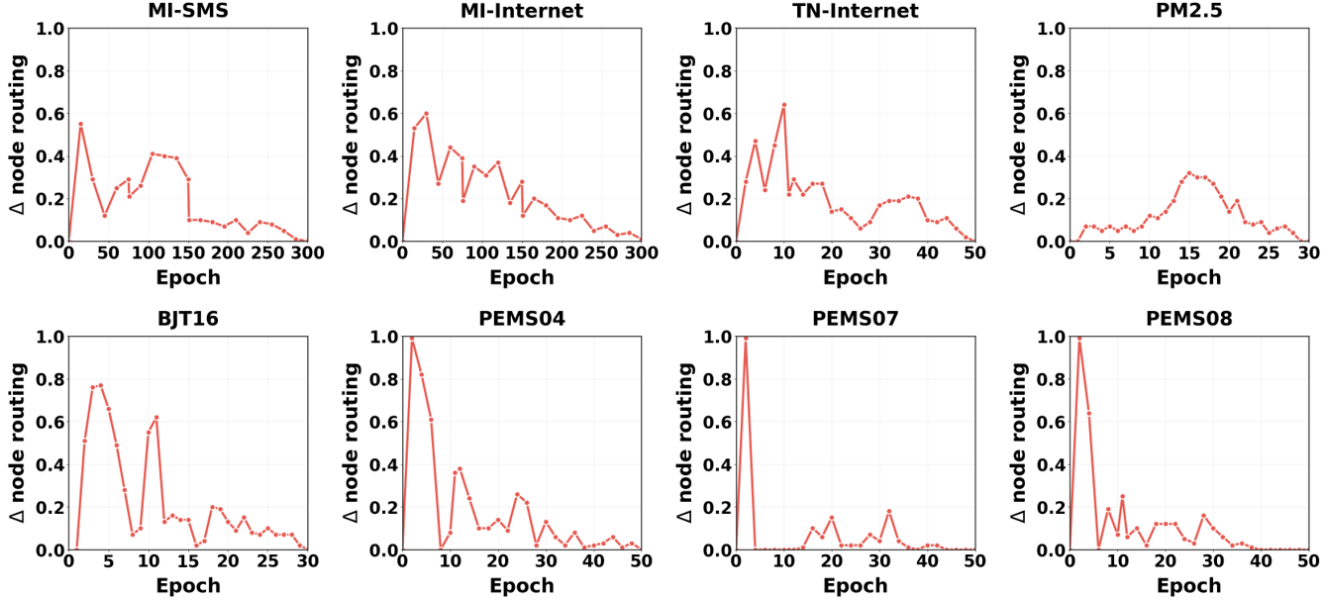


Figure 3: Fraction of spatial units changing their dominant prototype during training.

5.4 Effect of BPA on Prototype Routing

We analyze the effect of Bin-Prototype Affinity on prototype routing. For this analysis, we compare the routing weights of the full model with those obtained after removing only the BPA term from the routing cost while keeping the trained parameters fixed. This setting isolates the effect of BPA on routing behavior from changes in the learned parameters. Route Chg. denotes the fraction of spatial units whose dominant routed prototype changes, and Routing JS denotes the Jensen–Shannon divergence between the two soft routing distributions [19]. Active Base/Full denotes the number of active prototypes without and with BPA, respectively.

Table 7: Effect of BPA on forecasting and prototype routing behavior.

Dataset	Full (RMSE)	w/o BPA (RMSE)	Route Chg.	Routing JS	Active Base/Full	BPA Ratio
MI-Internet	134.56	153.05	0.7776	0.2322	28 / 89	0.1030
TN-Internet	43.91	45.80	0.7358	0.0398	18 / 54	0.0419
PM2.5	9.91	10.05	0.6405	0.0516	20 / 48	0.0491
PEMS04	252.95	256.68	0.5933	0.1248	4 / 18	0.0898

Table 7 reports forecasting error and prototype routing statistics. Removing BPA increases the forecasting error compared with the full model on the reported datasets. It also changes the dominant routed prototype for 77.8%, 73.6%, 64.1%, and 59.3% of spatial units, respectively. The non-zero Routing JS values show that the soft routing distributions also change when the BPA term is removed from the routing cost. The number of active prototypes is larger in the full model than in the variant without BPA. These results indicate that, under the fixed-parameter comparison, frequency-bin-level affinity modifies the routing cost and is associated with both forecasting error and prototype routing behavior.

5.5 Prototype Routing and BPA Dynamics

We analyze whether BPA is involved in the formation of prototype routing during training. Since BPA introduces frequency-bin-level temporal responses into the routing cost, we examine the evolution of dominant prototype assignments and the learned BPA coefficient α .

We report routing change, prototype region IoU, and prototype centroid shift. Routing change measures the fraction of spatial units whose dominant prototype changes between recorded epochs. Region IoU and centroid shift measure the stability of the resulting prototype-induced regions.

Table 8: Prototype routing statistics and BPA coefficient at selected training epochs.

Dataset	Epoch	Rout. Chg.	Reg. IoU	Cent. Shift	α
MI-Internet	1	0.0000	1.0000	0.0000	0.395
	75	0.3925	0.9344	0.0092	0.750
	150	0.2800	0.9470	0.0058	0.807
	225	0.1250	0.9550	0.0020	0.874
	300	0.0075	0.9985	0.0000	0.897
PM2.5	1	0.0000	1.0000	0.0000	0.397
	8	0.0500	0.9515	0.0006	0.407
	15	0.3225	0.8647	0.0007	0.466
	23	0.0775	0.9596	0.0003	0.524
	30	0.0025	0.9995	0.0000	0.529
PEMS04	1	0.0000	1.0000	0.0000	0.386
	12	0.3616	0.9820	0.0000	0.433
	25	0.2573	0.9648	0.0000	0.483
	37	0.0782	0.9865	0.0000	0.516
	50	0.0000	1.0000	0.0000	0.524

Figure 3 and Table 8 show that prototype assignments change during training and become stable near convergence. In all three datasets, routing change is non-zero at intermediate epochs and approaches zero at the final epoch, while region IoU returns close to one. This result shows that the final prototype routing is obtained through learned reassignment rather than by preserving the initial assignments.

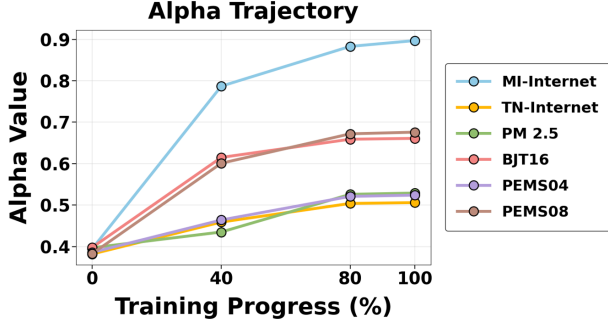


Figure 4: Training trajectory of the BPA coefficient α .

Figure 4 shows the trajectory of the learned BPA coefficient over normalized training progress. The coefficient increases during training across the reported datasets. For the three datasets included in Table 8, the coefficient increases from 0.395 to 0.897 on MI-Internet, from 0.397 to 0.529 on PM2.5, and from 0.386 to 0.524 on PEMS04. Since α weights BPA in the routing cost, this result indicates that the frequency-bin-level affinity term remains active and is adjusted during optimization.

Together with Table 7, these results support the role of BPA as a routing component. Removing BPA from the trained model changes the routing assignments, while the training dynamics show that prototype routing evolves with a learned BPA contribution. Thus, frequency-bin-level temporal responses are incorporated into prototype routing through the BPA-based routing cost.

5.6 Performance on Spatial Units with Different Spectral Properties

We analyze forecasting results on spatial units with different spectral properties. This analysis examines whether the use of frequency-bin-level temporal responses is related to forecasting results for spatial units with different temporal-frequency properties. Since average dataset-level errors do not separate periodicity-dominant units from mixed-frequency units, we group spatial units by their spectral properties and evaluate them separately.

Periodicity-dominant units are selected as the top 20% spatial units according to the 24-step autocorrelation-based periodicity score. Mixed-frequency units are selected as the top 20% spatial units according to a score combining spectral entropy and band-energy balance. Dom. Power Err. denotes the absolute error between the predicted and ground-truth dominant-frequency power ratios and measures how well the dominant spectral component is preserved. The periodicity-based selection criterion identifies units with strong daily recurrence, while Dom. Power Err. evaluates whether the dominant spectral component of these units

is preserved in the prediction. Δ RMSE is computed on first-order temporal differences and measures the error of predicted temporal changes.

Table 9: Forecasting results on periodicity-dominant and mixed-frequency spatial units.

Dataset	Model	Periodicity-dominant	Mixed-frequency		
		Dom. Power Err. ↓	RMSE	MAE	Δ RMSE
MI-Internet	BPR-Net (Ours)	0.0255	116.75	63.15	161.95
	TimeFilter	0.0325	118.48	64.19	168.01
	GLSTTN	0.0326	121.90	70.73	164.74
PM2.5	BPR-Net (Ours)	0.0108	10.11	4.72	14.05
	TimeXer	0.0223	11.44	5.33	14.73
	DRFormer	0.0161	10.26	4.78	14.31
PEMS08	BPR-Net (Ours)	0.0026	267.87	144.70	364.56
	DRFormer	0.0034	293.24	173.28	390.94
	MTGNN	0.0099	328.53	178.61	438.43

Table 9 reports results for periodicity-dominant and mixed-frequency spatial units. On periodicity-dominant units, BPR-Net reports lower Dom. Power Err. values across the selected datasets, indicating a smaller discrepancy in the dominant spectral component. On mixed-frequency units, BPR-Net reports lower RMSE, MAE, and Δ RMSE than the compared baselines. These results show that the reported errors are lower for both periodicity-dominant and mixed-frequency spatial units.

This analysis is connected to the bin-resolved design of BRSE. Periodicity-dominant units examine whether the model preserves dominant spectral components, while mixed-frequency units examine cases where temporal energy is distributed across multiple frequency components. The results indicate that frequency-bin-level temporal responses are associated with forecasting results for spatial units with different spectral properties. Since BPA uses these responses in the routing cost, this analysis provides complementary evidence that the bin-level temporal representation is relevant to the prototype-based spatial relation learning component.

Table 10: Computational cost and forecasting error on MI-Internet and PM2.5. Trainable parameters are reported in millions, training time per epoch and inference time are reported in seconds, and memory usage is reported in MB.

Dataset	Model	Params (M)	Train/Epoch (s)	Infer (s)	Memory (MB)	MAE
MI-Internet	TimeFilter	0.71	128.51	10.400	5969.80	78.00
	GLSTTN	1.20	3.81	0.237	1409.60	84.36
	TimeXer	8.60	18.15	0.898	1213.95	78.53
	DRFormer	0.27	5.86	0.017	114.80	79.83
	SimpleTM	0.41	0.80	0.0276	12.80	91.84
	RAST	0.27	4.90	0.019	521.18	103.93
	BPR-Net (Ours)	1.25	1.79	0.016	1414.43	76.57
PM2.5	DRFormer	0.75	8.90	0.840	4432.80	5.38
	TimeXer	0.12	6.34	0.440	1242.74	5.74
	TimeFilter	0.35	195.17	33.840	5957.46	5.61
	TAU	86.43	5.18	0.350	3359.20	6.25
	SimpleTM	0.14	2.00	0.120	73.03	6.95
	MTGNN	7.90	26.10	0.920	1814.60	6.08
	BPR-Net (Ours)	1.25	3.17	0.108	1434.50	5.18

5.7 Efficiency and Hyperparameter Analysis

5.7.1 Computational Efficiency Analysis. Table 10 reports forecasting error and computational cost on MI-Internet and PM2.5. We compare trainable parameters, training time per epoch, inference time, memory usage, and MAE. On both datasets, BPR-Net reports the lowest MAE among the compared models and the shortest inference time. SimpleTM reports lower memory usage and shorter training time per epoch, while its MAE is higher than that of BPR-Net in the reported settings.

On MI-Internet, BPR-Net reports an MAE of 76.57 with an inference time of 0.016 seconds. On PM2.5, it reports an MAE of 5.18 with an inference time of 0.108 seconds. These results describe the position of BPR-Net in terms of forecasting error, inference time, and memory usage in the two reported settings.

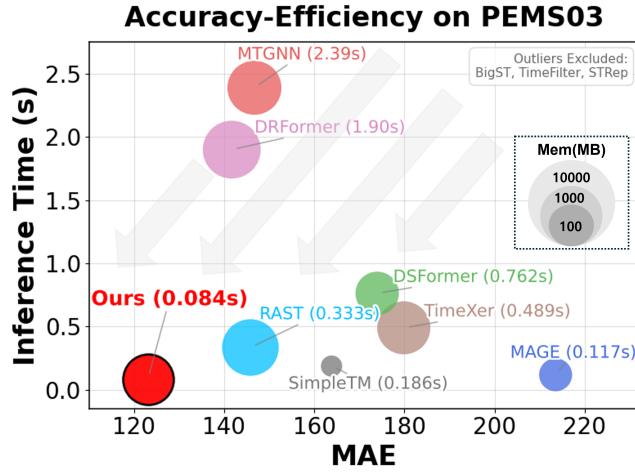


Figure 5: Inference time, MAE, and memory usage on PEMS03. Bubble size indicates memory usage.

In addition to the two tabulated grid-based settings, we visualize the efficiency–accuracy trade-off on the node-based PEMS03 dataset. Figure 5 compares inference time, MAE, and memory usage on PEMS03. BPR-Net is located in a region with lower error and shorter inference time than most baselines, while its memory usage remains within a moderate range.

5.7.2 Hyperparameter Analysis. We analyze two routing hyperparameters, ϵ and τ , together with the initialization of the learnable BPA coefficient α_{init} . The ratio τ/ϵ determines the effective sharpness of the routing distribution; we vary ϵ and τ separately while holding the other fixed to examine sensitivity around the default operating point. The parameter α_{init} specifies the initial contribution of BPA before optimization. Table 11 reports RMSE and MAE on MI-Internet, PM2.5, and PEMS03 under different settings.

The forecasting errors vary within a limited range under the tested settings. The default setting, $\epsilon = 0.08$, $\tau = 1.4$, and $\alpha_{\text{init}} = 0.5$, is among the lowest-error settings in most cases. The changes are more visible on MI-Internet, especially for ϵ and τ , while PM2.5 and PEMS03 show smaller differences across the tested settings.

Table 11: Effect of routing hyperparameters ϵ and τ , and BPA coefficient initialization α_{init} , on forecasting error.

Hyperparam.	Value	MI-Internet		PM2.5		PEMS03	
		RMSE	MAE	RMSE	MAE	RMSE	MAE
ϵ	0.04	139.73	77.25	9.94	5.20	215.16	123.60
	0.08	134.56	76.57	9.91	5.18	214.66	123.08
	0.12	138.82	77.28	9.92	5.21	215.74	123.14
τ	1.2	136.05	78.19	9.93	5.19	214.72	122.68
	1.4	134.56	76.57	9.91	5.18	214.66	123.08
	1.6	138.44	81.24	9.92	5.21	215.80	123.24
α_{init}	0.3	135.40	76.90	9.94	5.20	215.53	123.21
	0.5	134.56	76.57	9.91	5.18	214.66	123.08
	0.7	135.78	76.22	9.91	5.21	215.54	123.09

The results for α_{init} show limited variation within the tested range. This indicates that the reported forecasting errors are not restricted to a single initialization of the learnable BPA coefficient.

6 Conclusion

This paper presented BPR-Net, a frequency-bin-aware prototype routing model for spatiotemporal forecasting. BPR-Net uses frequency-bin-level temporal responses as a criterion for prototype-based spatial relation learning. BRSE represents each spatial unit through bin-level temporal responses, and BPA incorporates these responses into the routing cost. The spectral signature is used for frequency-aware gating of the routed representation.

Experiments on nine datasets across mobile traffic, air quality, taxi flow, and traffic sensor domains show lower forecasting errors in most settings compared with 19 baselines. Ablation studies and additional analyses indicate that frequency-bin-level responses are associated with forecasting performance, prototype routing behavior, and results for spatial units with different spectral properties. These results support the connected design of BPR-Net, where bin-level temporal responses are constructed by BRSE, incorporated into the routing cost through BPA, and used for frequency-aware modulation through the spectral signature.

The current evaluation focuses on one-step-ahead forecasting. Future work will examine multi-step forecasting and further analyze the learned prototype assignments across different forecasting horizons.

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GenAI Usage Disclosure

The authors used ChatGPT by OpenAI for language editing, grammar checking, assistance in manuscript organization, and limited

assistance with code formatting and repository preparation. All experimental results, data processing procedures, and technical claims were verified by the authors. Generative AI tools were not used to fabricate data, generate unverified results, or make autonomous research decisions.

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